AISC5020
Week 8
Bayesian Statistics and Bayesian Networks

Presented by

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Chapter Outline

- Bayes' rule, priors, posteriors, conjugate families
- Bayesian networks: structure, inference, factorization, conditional independence
- DAGs, d-separation, local marginals

Bayes' Rule

- Suppose we have observed data $y = (y_1, y_2, \ldots, y_n)$.
- Define the prior distribution of θ as $p(\theta)$ and the sampling distribution as $p(y|\theta)$. The posterior distribution of θ given data y is $p(\theta|y) \propto p(y|\theta)p(\theta)$.
- The posterior prediction of a new y_* assuming y and y_* are conditional independence given θ is

$$p(y_*|y) = \int p(y_*, \theta|y)d\theta = \int p(y_*|\theta, y)p(\theta|y)d\theta = \int p(y_*|\theta)p(\theta|y)d\theta$$

- Suppose θ_1 and θ_2 are candidate parameters. Their posterior likelihood ratio or odds is defined as

$$\frac{p(\theta_1|y)}{p(\theta_2|y)} = \frac{p(y|\theta_1)p(\theta_1)}{p(y|\theta_2)p(\theta_2)}$$

Bayes' Rule

- Example (Coin Toss)
- Data $y = (y_1, y_2, \dots, y_n)$, where $y_i \in \{0,1\}$
- Let $\theta = P(Y = 1) \sim Beta(1,1)$
- The sampling distribution $P(y|\theta) = \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})}$
- $P(\theta|y) \propto P(y|\theta)P(\theta) = \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})} \Longrightarrow \theta|y \sim Beta\big(1+n\bar{y}, 1+n(1-\bar{y})\big)$
- For a new $y_* \in \{0,1\}$,

$$p(y_*|y) = \int_0^1 p(y_*|\theta)p(\theta|y)d\theta = \frac{\Gamma(n+2)}{\Gamma(1+n\bar{y})\Gamma\big(1+n(1-\bar{y})\big)} \int_0^1 \theta^{y_*}(1-\theta)^{1-y_*}\,\theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})}d\theta$$

$$= \frac{\Gamma(n+2)}{\Gamma(1+n\bar{y})\Gamma\big(1+n(1-\bar{y})\big)} \int_0^1 \theta^{n\bar{y}+y_*}(1-\theta)^{n(1-\bar{y})+1-y_*}d\theta$$

$$= \frac{\Gamma(n+2)}{\Gamma(1+n\bar{y})\Gamma\big(1+n(1-\bar{y})\big)} \frac{\Gamma(n\bar{y}+y_*+1)\Gamma(n+2-n\bar{y}-y_*)}{\Gamma(n+3)}$$

$$p(y_*=1|y) = \frac{1+n\bar{y}}{n+2}, p(y_*=0|y) = \frac{1+n(1-\bar{y})}{n+2}$$

Prior

- How priors are chosen?
  - The prior represents a distribution of all plausible values of θ in a population. The true parameter θ is viewed as randomly drawn from the population.
  - It does not need to be very accurate because, with enough data, the likelihood dominates the prior.
  - A prior is simply a way to encode what you believe about θ before seeing data — and sometimes "knowing nothing" is modeled as giving all values equal weight (with caution).
- Uniform prior is not invariant under transformation.
  - Suppose person A wants to estimate a probability $\theta \in (0,1)$ and person B wants to estimate $\phi = \theta^2$.
  - Both persons assume the same prior, that are $\theta \sim \text{Uniform}(0,1)$ and $\phi \sim \text{Uniform}(0,1)$.
  - Since $\frac{d\phi}{d\theta} = 2\theta$, $p(\theta) = p_\phi(\theta^2)\left|\frac{d\phi}{d\theta}\right| = 2\theta \neq \text{Uniform}(0,1)$.
  - Since they produced different priors, they will get different conclusions.

Jeffrey Prior

- An invariant prior is a prior distribution that does not change when we reparametrize the model.
- If you describe the same problem using different parameters, an invariant prior gives consistent results, regardless of how you express the parameter.
- A prior $p(\theta)$ is **invariant** if, under a transformation $\phi = g(\theta)$, it transforms correctly, that is

$$p(\phi)\, d\phi = p(\theta)\, d\theta$$

- Define the Fisher information for a **single observation** $I(\theta) = E\left(\left(\frac{\partial \ln p(y|\theta)}{\partial \theta}\right)^2\right) = E\left(-\frac{\partial^2 \ln p(y|\theta)}{\partial \theta^2}\right)$

  Proof.

$$E\left[\frac{\partial^2 \log p(y|\theta)}{\partial \theta^2}\right] = E\left[\frac{1}{p(y|\theta)}\frac{\partial^2 p(y|\theta)}{\partial \theta^2} - \left(\frac{1}{p(y|\theta)}\frac{\partial p(y|\theta)}{\partial \theta}\right)^2\right]$$

$$= \int \left\{\frac{1}{p(y|\theta)}\frac{\partial^2 p(y|\theta)}{\partial \theta^2} - \left(\frac{1}{p(y|\theta)}\frac{\partial p(y|\theta)}{\partial \theta}\right)^2\right\} p(y|\theta) dy = \int \frac{\partial^2 p(y|\theta)}{\partial \theta^2} dy - E\left[\left(\frac{\partial \log p(y|\theta)}{\partial \theta}\right)^2\right]$$

$$= \frac{\partial^2}{\partial \theta^2} \int p(y|\theta)\, dy - I(\theta) = -I(\theta).$$

Jeffrey Prior

- The most common invariant prior is the **Jeffreys prior**: $p(\theta) \propto \sqrt{I(\theta)}$ where $I(\theta)$ is the **Fisher Information**.
- Rework our example.
- Data $y = (y_1, y_2, \ldots, y_n)$, where $y_i \in \{0,1\}$
- Let $\theta = P(Y = 1)$
- The sampling distribution $P(y|\theta) = \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})}$

n observations.

$$\frac{d \ln p(y|\theta)}{d\theta} = \frac{y}{\theta} - \frac{1-y}{1-\theta}$$

single observation

$$\frac{d^2 \ln p(y|\theta)}{d\theta^2} = -\frac{y}{\theta^2} - \frac{1-y}{(1-\theta)^2}$$

$$I(\theta) = E\left(-\frac{d^2 \ln p(y|\theta)}{d\theta^2}\right) = \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)}$$

Jeffrey Prior

- Data $y = (y_1, y_2, \dots, y_n)$, where $y_i \in \{0,1\}$
- Let $\theta = P(Y = 1) \sim p(\theta) \propto \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}$
- The sampling distribution $P(y|\theta) = \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})}$
- $P(\theta|y) \propto P(y|\theta)P(\theta) = \theta^{n\bar{y}-\frac{1}{2}}(1-\theta)^{n(1-\bar{y})-\frac{1}{2}} \Longrightarrow \theta|y \sim Beta\left(n\bar{y}+\frac{1}{2}, n(1-\bar{y})+\frac{1}{2}\right)$
- For a new $y_* \in \{0,1\}$,

$$p(y_*|y) = \int_0^1 p(y_*|\theta)p(\theta|y)d\theta = \frac{\Gamma(n+1)}{\Gamma\left(n\bar{y}+\frac{1}{2}\right)\Gamma\left(n(1-\bar{y})+\frac{1}{2}\right)}\int_0^1 \theta^{y_*}(1-\theta)^{1-y_*}\,\theta^{n\bar{y}-\frac{1}{2}}(1-\theta)^{n(1-\bar{y})-\frac{1}{2}}d\theta$$

$$= \frac{\Gamma(n+1)}{\Gamma\left(n\bar{y}+\frac{1}{2}\right)\Gamma\left(n(1-\bar{y})+\frac{1}{2}\right)}\int_0^1 \theta^{n\bar{y}+y_*-\frac{1}{2}}(1-\theta)^{n(1-\bar{y})+1-y_*-\frac{1}{2}}d\theta$$

$$= \frac{\Gamma(n+1)}{\Gamma\left(n\bar{y}+\frac{1}{2}\right)\Gamma\left(n(1-\bar{y})+\frac{1}{2}\right)}\frac{\Gamma\left(n\bar{y}+y_*+\frac{1}{2}\right)\Gamma\left(n+\frac{3}{2}-n\bar{y}-y_*\right)}{\Gamma(n+2)}$$

$$p(y_*=1|y) = \frac{n\bar{y}+\frac{1}{2}}{n+1}, p(y_*=0|y) = \frac{n(1-\bar{y})+\frac{1}{2}}{n+1}$$

Conjugate Prior

- Suppose that $\mathcal{P}$ is a class of prior distributions for θ. A prior distribution is conjugate if $p(\theta|y) \in \mathcal{P}$, that is the posterior has the same form as the prior.
- Example. Binomial likelihood and Beta prior → Posterior is also Beta.
- Advantages
  - Gives closed-form (analytic) solutions
  - Easy to compute and interpret
  - Useful building blocks for more complex models
- For exponential family $p(y|\theta) = f(y)g(\theta)e^{\phi(\theta)'u(y)}$,
- the natural conjugate prior is $p(\theta) \propto g(\theta)^{\eta} e^{\phi(\theta)'\nu}$.

$$p(\theta|y) \propto g(\theta)^{\eta} e^{\phi(\theta)'\nu} \prod_{j=1}^{n} f(y_j) g(\theta) e^{\phi(\theta)'u(y_j)} \propto g(\theta)^{\eta+n} e^{\phi(\theta)'\left(\nu+\sum_{j=1}^{n} u(y_j)\right)}$$

Conjugate Prior

- Example. Suppose $y = (y_1, y_2, \dots, y_n)$, where $y_i \in \{1,2, \dots, K\}$ (There are K categories).
- Parameter $\theta = (\theta_1, \dots, \theta_K)$ with $\sum_{j=1}^{K} \theta_j = 1$.
- Likelihood is Multinomial $p(y|\theta) \propto \prod_{j=1}^{K} \theta_j^{n_j}$, where $n_k = \sum_{j=1}^{n} I(y_j = k)$.
- The prior for $\theta \sim$Dirichlet$(\alpha_1, \alpha_2, \dots, \alpha_K)$ is $p(\theta) \propto \prod_{j=1}^{K} \theta_j^{\alpha_j - 1}$, where $\alpha_k > 0$ are positive numbers.
- The posterior for θ is $p(\theta|y) \propto \prod_{j=1}^{K} \theta_j^{n_j + \alpha_j - 1}$. Thus, $\theta|y \sim$Dirichlet$(n_1 + \alpha_1, \dots, n_K + \alpha_K)$.

Conjugate Prior

- The predictive distribution for a new observation $y_* \in \{1,2,\dots,K\}$ is

$$p(y_* = k|y) = \frac{\Gamma\left(n + \sum_{j=1}^{K} \alpha_j\right)}{\prod_{j=1}^{K} \Gamma(n_j + \alpha_j)} \int_{[0,1]^K} \theta_k \prod_{j=1}^{K} \theta_j^{n_j + \alpha_j - 1} \, d\theta_1 \cdots d\theta_K$$

$$= \frac{\Gamma\left(n + \sum_{j=1}^{K} \alpha_j\right)}{\prod_{j=1}^{K} \Gamma(n_j + \alpha_j)} \int_{[0,1]^K} \prod_{\substack{j=1 \\ j \neq k}}^{K} \theta_j^{n_j + \alpha_j - 1} \, \theta_j^{n_k + \alpha_k + 1 - 1} d\theta_1 \cdots d\theta_K$$

$$= \frac{\Gamma\left(n + \sum_{j=1}^{K} \alpha_j\right)}{\prod_{j=1}^{K} \Gamma(n_j + \alpha_j)} \cdot \frac{\prod_{\substack{j=1 \\ j \neq k}}^{K} \Gamma(n_j + \alpha_j) \, \Gamma(n_k + \alpha_k + 1)}{\Gamma\left(n + 1 + \sum_{j=1}^{K} \alpha_j\right)} = \frac{n_k + \alpha_k}{n + \sum_{j=1}^{K} \alpha_j}$$

where $n = \sum_{j=1}^{K} n_j$.

Conjugate Prior

- Example. Normal distribution with known variance
- $p(y|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\theta)^2}$ and $p(\theta) \propto e^{-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2}$
- $p(\theta|y) \propto e^{-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2} \prod_{j=1}^{n} e^{-\frac{1}{2\sigma^2}(y_j-\theta)^2} = e^{-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2 - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\theta)^2} =$

  $e^{-\frac{1}{2\tau_0^2}(\theta^2-2\mu_0\theta+\mu_0^2)-\frac{1}{2\sigma^2}\left(n\theta^2-2\theta\sum_{j=1}^{n}y_j+\sum_{j=1}^{n}y_j^2\right)} = e^{-\frac{1}{2}\left(\left(\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}\right)\theta^2-2\left(\frac{\mu_0}{\tau_0^2}+\frac{n\bar{y}}{\sigma^2}\right)\theta+\frac{\mu_0^2}{\tau_0^2}+\frac{\sum_{j=1}^{n}y_j^2}{\sigma^2}\right)} \propto$

  $e^{-\frac{1}{2\tau_1^2}(\theta-\mu_1)^2}$

- where $\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$ and $\mu_1 = \frac{\frac{\mu_0}{\tau_0^2}+\frac{n\bar{y}}{\sigma^2}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}}$.

Conjugate Prior

- The predictive distribution for a new observation y_* is

$$p(y_*|y) = \int_{-\infty}^{\infty} p(y_*|\theta)p(\theta|y)d\theta$$

- Since $y_*|\theta \sim n(\theta, \sigma^2)$ and $\theta|y \sim n(\mu_1, \tau_1^2)$, $y_* = \theta + \varepsilon$, $\theta \sim n(\mu_1, \tau_1^2)$ and $\varepsilon \sim n(0, \sigma^2)$ and θ and ε are independent.
- $y_*|y \sim n(\mu_1, \sigma^2 + \tau_1^2)$.

$$E(y_*|y) = E(E(y_*|\theta)|y) = E(\theta|y) = \mu_1$$
$$Var(y_*|y) = Var(E(y_*|\theta)|y) + E(Var(y_*|\theta)|y) = Var(\theta|y) + E(\sigma^2|y) = \sigma^2 + \tau_1^2$$

Conjugate Prior

- Example. Normal distribution with known mean θ but unknown variance σ^2
- $p(y = (y_1, y_2, \dots, y_n)|\theta) \propto (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\theta)^2}$
- Prior for σ^2 is $p(\sigma^2) \propto (\sigma^2)^{-(\alpha+1)} e^{-\frac{\beta}{\sigma^2}} = IG(\alpha, \beta)$
- $p(\sigma^2|y) \propto (\sigma^2)^{-(\alpha+1)} e^{-\frac{\beta}{\sigma^2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\theta)^2} = (\sigma^2)^{-\left(\frac{n}{2}+\alpha+1\right)} e^{-\frac{\beta}{\sigma^2}-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\theta)^2}$
- Thus, $\sigma^2|y \sim IG\left(\frac{n}{2}+\alpha, \beta+\frac{1}{2}\sum_{j=1}^{n}(y_j-\theta)^2\right)$

Conjugate Prior

- The predictive distribution for a new observation y_* is

$$p(y_*|y) = \int_0^\infty p(y_*|\sigma^2)p(\sigma^2|y)d\sigma^2$$

- Since $y_*|\sigma^2 \sim n(\theta, \sigma^2)$ and $\sigma^2|y \sim IG(a,b)$, where $a = \frac{n}{2} + \alpha$ and $b = \beta + \frac{1}{2}\sum_{j=1}^{n}(y_j - \theta)^2$,

$$p(y_*|y) \propto \frac{b^a}{\Gamma(a)} \int_0^\infty (\sigma^2)^{-\frac{1}{2}} e^{-\frac{1}{2\sigma^2}(y_*-\theta)^2} (\sigma^2)^{-(a+1)} e^{-\frac{b}{\sigma^2}} d\sigma^2$$

$$= \frac{b^a}{\Gamma(a)} \int_0^\infty (\sigma^2)^{-\left(a+\frac{3}{2}\right)} e^{-\frac{1}{\sigma^2}\left(b+\frac{1}{2}(y_*-\theta)^2\right)} d\sigma^2 = \frac{b^a}{\Gamma(a)} \cdot \frac{\Gamma\left(a+\frac{1}{2}\right)}{\left(b+\frac{1}{2}(y_*-\theta)^2\right)^{a+\frac{1}{2}}}$$

$$\propto \left(1 + \frac{1}{2b}(y_* - \theta)^2\right)^{-\left(a+\frac{1}{2}\right)}$$

Conjugate Prior

Since the density of a Student's t random variable with ν degrees of freedom, location μ, and scale s:

$$f(x) \propto \left[1 + \frac{1}{\nu}\left(\frac{x-\mu}{s}\right)^2\right]^{-\frac{(\nu+1)}{2}}, \qquad \left(1 + \frac{1}{2b}(y_* - \theta)^2\right)^{-\left(a+\frac{1}{2}\right)}$$

- the predictive distribution is a Student's t distribution. By solving $\nu = 2a$ and $2b = \nu s^2$. We have $\nu = 2a$ and $s^2 = \frac{b}{a}$.

$$y_* \mid y \sim t_{2a}\left(\theta, \sqrt{\frac{b}{a}}\right) = t_{n+2\alpha}\left(\theta, \sqrt{\frac{\beta + \frac{1}{2}\sum_{j=1}^{n}\left(y_j - \theta\right)^2}{\frac{n}{2} + \alpha}}\right)$$

Conjugate Prior

- Example. Poisson Model
- $p(y=(y_1,y_2,\dots,y_n)|\theta) \propto \theta^{\sum_{j=1}^{n} y_j} e^{-n\theta} = \theta^{n\bar{y}} e^{-n\theta}$, where $y_j = 0,1,2,\dots$
- Prior for θ is Gamma(α,β), that is $p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}, \theta > 0.$
- $p(\theta|y) \propto \theta^{\alpha-1}e^{-\beta\theta}\theta^{n\bar{y}}e^{-n\theta} = \theta^{\alpha+n\bar{y}-1}e^{-(\beta+n)\theta} \Longrightarrow \theta|y \sim \text{Gamma}(\alpha+n\bar{y}, \beta+n)$
- Since $y_*|\theta \sim \text{Poisson}(\theta)$ and $\theta|y \sim \text{Gamma}(\alpha+n\bar{y}, \beta+n)$,

$$p(y_*|y) = \frac{(\beta+n)^{\alpha+n\bar{y}}}{\Gamma(\alpha+n\bar{y})}\int_0^\infty \frac{\theta^{y_*}e^{-\theta}}{y_*!}\theta^{\alpha+n\bar{y}-1}e^{-(\beta+n)\theta}d\theta$$

$$= \frac{(\beta+n)^{\alpha+n\bar{y}}}{\Gamma(\alpha+n\bar{y})\Gamma(y_*+1)}\int_0^\infty \theta^{\alpha+n\bar{y}+y_*-1}e^{-(n+1+\beta)\theta}d\theta = \frac{(\beta+n)^{\alpha+n\bar{y}}}{\Gamma(\alpha+n\bar{y})\Gamma(y_*+1)}\cdot\frac{\Gamma(\alpha+n\bar{y}+y_*)}{(n+1+\beta)^{\alpha+n\bar{y}+y_*}}$$

$$= \binom{\alpha+n\bar{y}+y_*-1}{y_*}\left(\frac{\beta+n}{\beta+n+1}\right)^{\alpha+n\bar{y}}\left(\frac{1}{\beta+n+1}\right)^{y_*} \sim \text{NegBinomial}(\alpha+n\bar{y}, \beta+n)$$

Bayesian Network

- If we have n discrete random variables each with domain size d (possible values), the full joint distribution $P(X_1, \dots, X_n)$ requires storing d^n numbers – completely impractical for real problems.
- Bayesian networks address this problem by leveraging conditional probability. Instead of storing all probabilities in a single large table, they organize them into many smaller conditional probability tables (CPTs) connected by a directed acyclic graph (DAG) that represents how variables are related.
- Together, the graph structure and these local probability tables provide enough information to reconstruct and compute any probability that could have been obtained from the full joint distribution—while using far less space and computation.

Bayesian Network

A simple example is a **medical diagnosis Bayesian network**. Suppose we have four binary random variables, each with $d = 2$ possible values:

- F: Flu, $\{0, 1\}$
- V: Fever, $\{0, 1\}$
- C: Cough, $\{0, 1\}$
- T: Positive test, $\{0, 1\}$

Without a Bayesian network

The full joint distribution is $P(F, V, C, T)$. Since there are $n = 4$binary variables, it contains $2^4 = 16$ possible combinations, such as

$$P(F = 1, V = 1, C = 0, T = 1).$$

In general, n variables with d possible values require d^n joint probabilities.

Bayesian Network

Suppose our knowledge suggests that **flu causes fever, cough, and the test result**: $F \longrightarrow V, F \longrightarrow C, F \longrightarrow T$.

The DAG encodes the conditional independence assumption $V \perp C \perp T \mid F$.

Therefore, instead of representing the joint distribution directly, we factor it as

$$P(F, V, C, T) = P(T|C, V, F)P(C|V, F)P(V|F)P(F) = P(F)P(V \mid F)P(C \mid F)P(T \mid F)$$

using four small CPTs.

For example, suppose $P(F = 1) = 0.10$ and the CPTs contain probabilities such as

$$P(V = 1 \mid F = 1) = 0.80, P(V = 1 \mid F = 0) = 0.05,$$

$$P(C = 1 \mid F = 1) = 0.70, P(C = 1 \mid F = 0) = 0.10,$$

and

$$P(T = 1 \mid F = 1) = 0.90, P(T = 1 \mid F = 0) = 0.05.$$

$$P(F = 1, V = 1, C = 0, T = 1) = P(F = 1)P(V = 1 \mid F = 1)P(C = 0 \mid F = 1)P(T = 1 \mid F = 1).$$

Since $P(C = 0 \mid F = 1) = 1 - 0.70 = 0.30$, we get

$$P(F, =, 1, V = 1C = 0T = 1) = (0.10)(0.80)(0.30)(0.90) = \boxed{0.0216}.$$

Bayesian Network

- A Bayesian network is a pair (G, P) where
  - G is a directed acyclic graph (DAG) with one node per random variable.
  - For each node X with parents Π (the set of nodes with directed edges to X), we specify a conditional probability table (CPT) (or conditional probability distribution) $P(X|\Pi)$.
- The graph G encodes **conditional independence assertions** (via the Markov condition / splitting property), and the CPTs give the local probabilities.
- Each variable is conditional independent of its non-descendants **given its parents**.

Consider the DAG: $A \to B, A \to C, B \to D$.

For variable B:

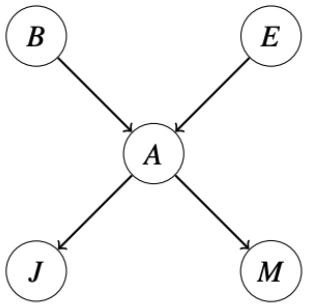
- Parent of B: A
- Descendant of B: D
- Non-descendant of B: C

The local Markov property says $B \perp C \mid A$.

Bayesian Network

- For example, A → B → C. We can infer from the DAG that C and A are conditional independent given B.
- **Factorization theorem** : $P(X_1, \dots, X_n) = \prod_{j=1}^{n} P\left(X_j \middle| parents\left(X_j\right)\right)$
- Example: Alarm Network. Variables: Burglary (B), Earthquake (E), Alarm (A), John calls (J), Mary calls (M).
  - Assume that the alarm can be triggered by a burglary or an earthquake, and Mary and John call if they hear it.
  - CPTs stored: $P(B), P(E), P(A|B,E), P(J|A) and\ P(M|A)$.
  - The joint probability factorizes into

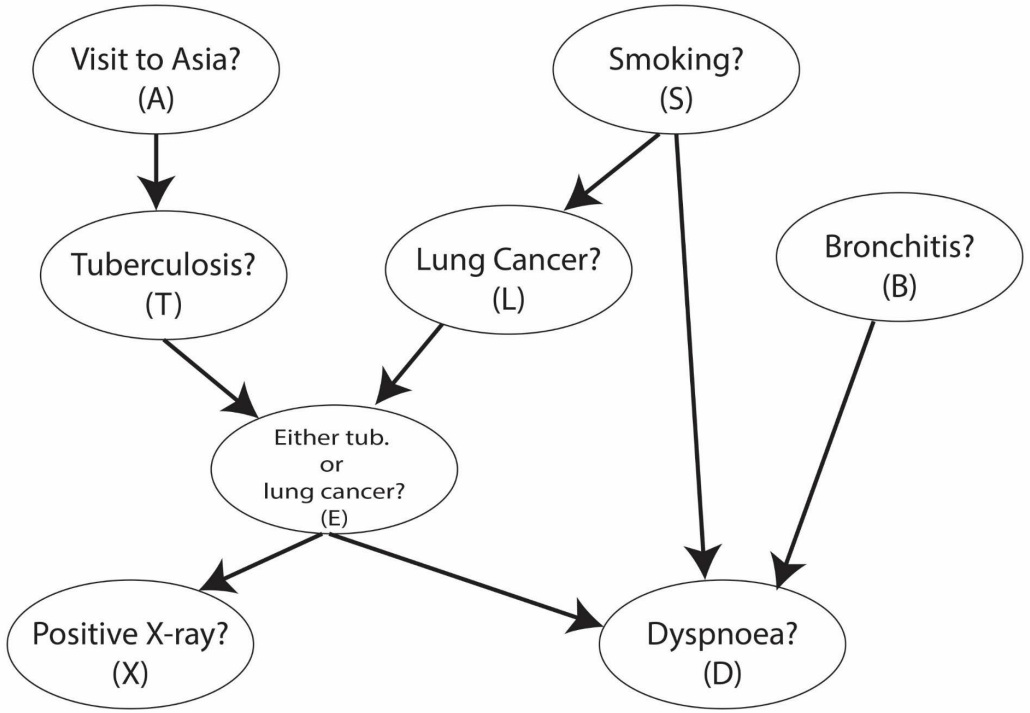
$$P(B,E,A,J,M) = P(B) \cdot P(E) \cdot P(A|B,E) \cdot P(J \mid A) \cdot P(M \mid A)$$



Bayesian Network

- Example: Suppose we want to determine whether a patient has tuberculosis, lung cancer, or bronchitis, based on possible causes (like smoking or travel to Asia) and symptoms (such as a positive chest X-ray or difficulty breathing).
- The joint probability of the DAG is given by

$$P(A,S,T,L,B,E,X,D) = P(X|E)P(D|E,S,B)P(E|T,L)P(T|A)P(L|S)P(A)P(S)P(B)$$



Bayesian Network

- In probability, if two variables are dependent, information flows between them. If they are conditionally independent, that flow of information has been blocked.
- Consider DAG: $A \rightarrow B \rightarrow C \rightarrow D$. Parent of $D = C$. Ancestors of $D = A, B, C$.
- Local Markov Property:
  - A node is conditionally independent of its ancestor nodes (and any other non-descendants) given its immediate parents. $P(D|C) = P(D|A, B, C)$.

    In general, $P(var|Parents(var), Ancestors(var)) = P(var|Parents(var))$.
  - If you do not know the value of C, then knowing A or B gives you a hint about what C might be, which in turn gives you a hint about D.
  - However, the moment you know the value of C, the pipeline is blocked. C holds all the up-to-date information needed to predict D.

Bayesian Network

- The Markov Blanket Property
  - A node is conditionally independent of every other variable in the entire network given its Markov Blanket.

    For a node X, its Markov blanket includes

    - Parents of X: knowing the parents blocks information from travelling downward to X's ancestors into X.

    - Children of X: By conditioning on X's children, we lock their values in place, blocking information from travelling up from further down the chain. For example, $X \rightarrow Y \rightarrow Z \Longrightarrow Z \rightarrow Y \rightarrow X$ (observing Z tells us something about X). However, given Y, observing Z provides no additional information about X or the path is blocked.

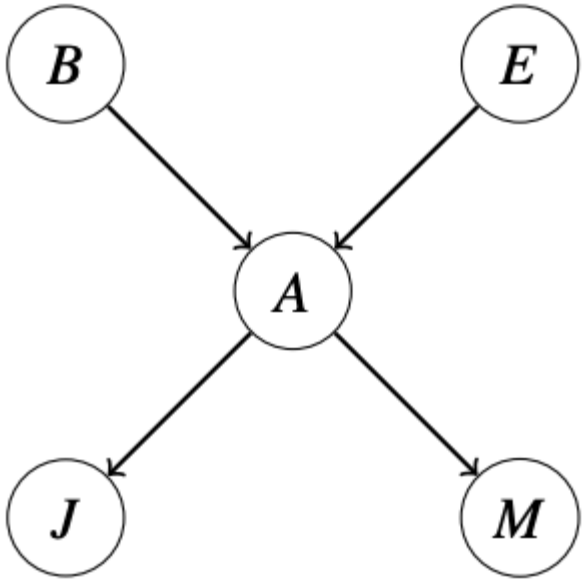
    - Co-parents of X (Other parents of X's children)

      - If two parents share a child ($X \rightarrow$ Child $\leftarrow$ Co-parent), they are independent. However, the moment we know the value of the child, the two parents suddenly become dependent on each other.

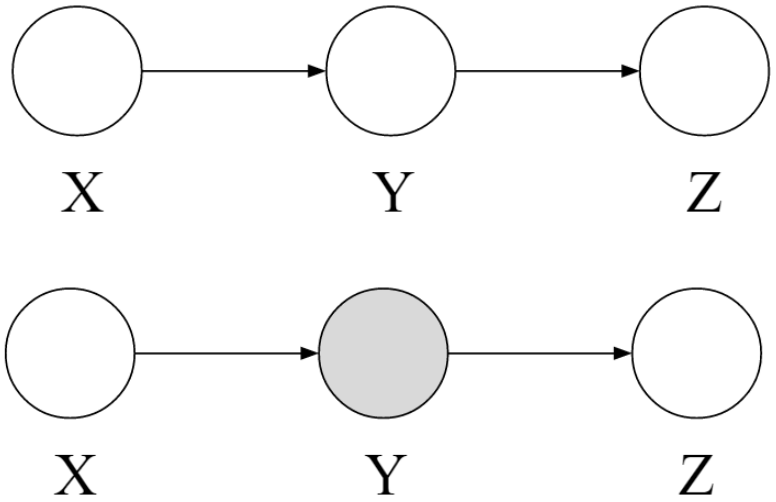
      - Because we have conditioned on X's children, we accidentally open up a pathway between X and its co-parents.

Bayesian Network

- Example. The Alarm Network
  - For node A: Parents (B, E), Children (J, M), Other parents (None). Markov Blanket of A is {B, E, J, M}.
  - For node J: Parents (A), Children (None), Other parents (None). Markov Blanket of J is {A}.
  - For node B: Parents (None), Children (A), Other parents (E). Markov Blanket of B is {A, E}.
  - A is independent of everything else given {B, E, J, M}.
  - J is independent of {B, E, M, everything else} given {A}.
  - B is independent of {J, M, everything else} given {A, E}.



Bayesian Network

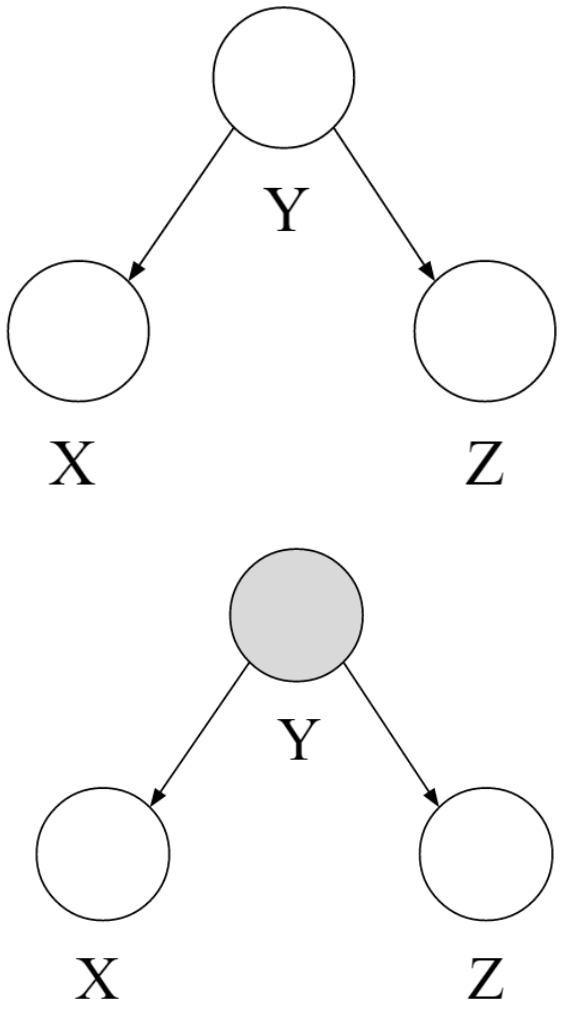


Chain

- Y is not observed. X and Z are not necessarily independent.
- $P(X,Y,Z) = P(Z|Y)P(Y|X)P(X)$
- Note $P(Z|Y,X) = P(Z|Y)$
- $P(X|Z,Y) = P(X|Y)$ X and Z are conditional independent given Y.

$$P(X|Z,Y) = \frac{P(X,Y,Z)}{P(Y,Z)} = \frac{P(Z|Y)P(Y|X)P(X)}{\sum_x P(Y|X=x)P(Z|Y)}$$
$$= \frac{P(Y|X)P(X)}{\sum_x P(Y|X=x)} = P(X|Y)$$

Bayesian Network

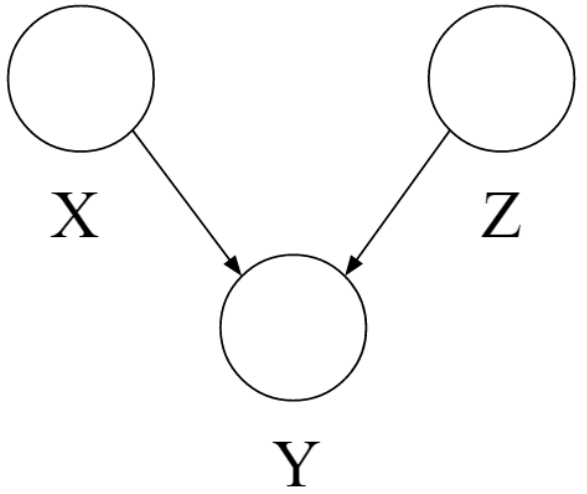


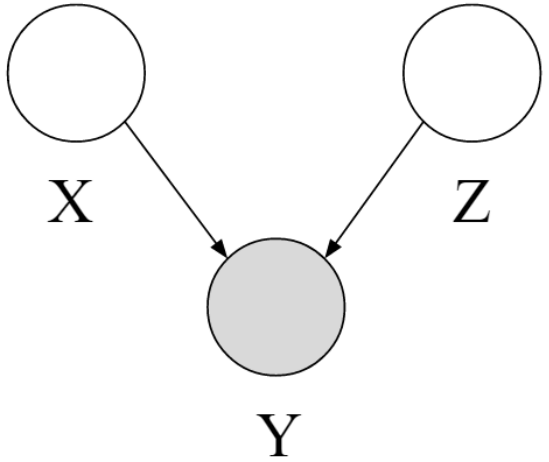
Common Causes (Fork)

- Y is not observed. X and Z are not necessarily independent.
- $P(X,Y,Z) = P(X|Y)P(Z|Y)P(Y)$
- $P(X|Z,Y) = P(X|Y)$ X and Z are conditional independent given Y.

$$P(X|Z,Y) = \frac{P(X,Y,Z)}{P(Y,Z)} = \frac{P(X|Y)P(Z|Y)P(Y)}{P(Y,Z)} = P(X|Y)$$

Bayesian Network

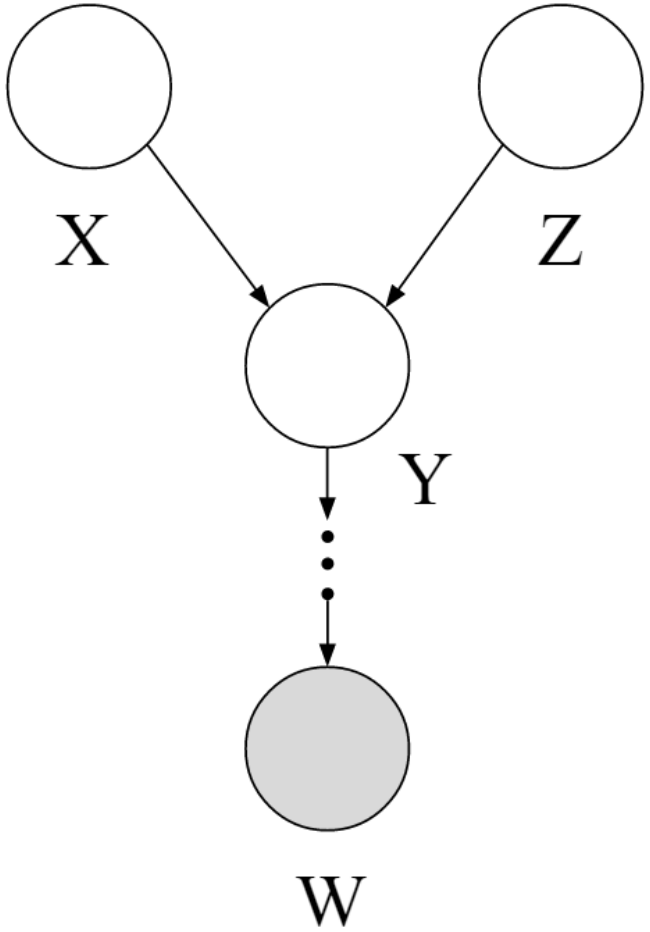




Common Effect (Collider)

- Y is not observed. X and Z are independent.
- $P(X,Y,Z) = P(Y|X,Z)P(X)P(Z)$
- But X and Z are not conditionally independent given Y.
- Suppose $P(X=1) = P(X=0) = 0.5$
- Suppose $P(Z=1) = P(Z=0) = 0.5$
- X and Z are independent.
- Suppose $P(Y=1|X=Z) = 1$ and $P(Y=0|X \neq Z) = 1$. That is $Y = I(X=Z)$.
- Given $Y = y \in \{0,1\}$, knowing X tells us the value of Z. So, X and Z are not conditionally independent given Y.

Bayesian Network



Another Common Effect (Collider)

- Y is not observed. X and Z are independent.
- W is a descendant of Y
- X and Z are not conditionally independent given Y.
- X and Z are not conditionally independent given W.

Bayesian Network

- Given a Bayesian network G, two nodes X and Y and a set of observed variables $Z = \{Z_1, \dots, Z_k\}$.
- If X and Y are **d-separated** by Z, then in every possible probability distribution that can be represented by the network, the conditional independence holds

$$X \text{ and } Y \text{ are conditionally independent given } Z = \{Z_1, \dots, Z_k\}$$

- The Core Idea: Active and Blocked Paths
  - Think of an undirected path between X and Y as a pipeline. Information can flow through this pipeline to create a statistical dependence between X and Y.
  - If every single path between X and Y is blocked ("inactive"), then X and Y are d-separated, meaning that they are guaranteed to be conditionally independent $(X \perp Y|Z)$.
  - If even one path allows information to flow freely from end to end (it is "active"), X and Y are d-connected, meaning they might be dependent.
  - To see if a path is blocked, we inspect its 3-node segments: the overlapping triples. If a single triple along a path is inactive (blocked), the entire path is inactive.

Bayesian Network

- The d-separation algorithm
  - Shade all observed nodes $Z = \{Z_1, \dots, Z_k\}$ in the graph.
  - Treat edges in the graph as undirected. Enumerate all undirected paths from X to Y.
  - For each such path,
    - Break it into overlapping triples (every three consecutive nodes along the path).
    - Check whether every triple on the path is active (dependence flow)
    - If a path contains a Chain or a Fork where the middle node is shaded $\Longrightarrow$ Path is blocked.
    - If a path contains a Collider $(\mathrm{A} \rightarrow \mathrm{B} \leftarrow \mathrm{C})$ where the middle node B (and all its downstream descendants) is unshaded $\Longrightarrow$ Path is blocked.

Bayesian Network

- The d-separation algorithm (Continue)
  - Two Cases:
    - If at least one path has all its triples active, the path is active. X and Y are d-connecting given Z and the conditional independence between X and Y given Z is not guaranteed.
    - If every path is inactive (blocked), then X and Y are d-separated by Z. The conditional independence between X and Y given Z holds in every probability distribution consistent with G.
  - To determine a triple active or inactive: Any path can be decomposed into overlapping triples of the form A – B – C.

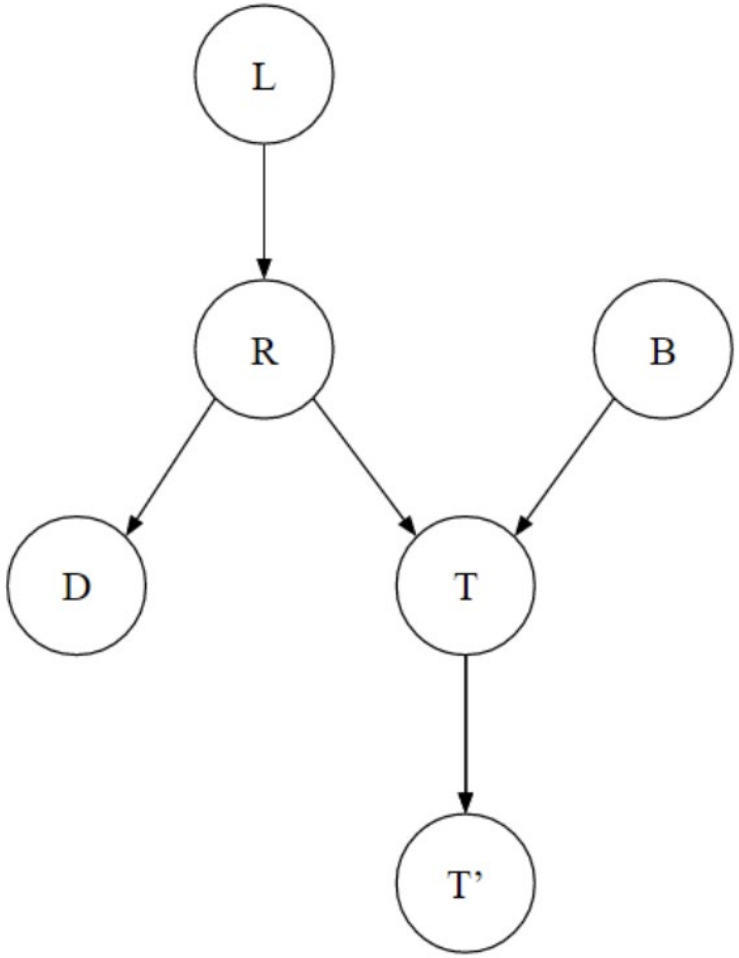
Bayesian Network

- The d-separation algorithm (Continue)

Triple Structure	Name	Middle Node B observed	Status of the triple
A→B→C	Causal Chain	Not observe	Active (dependence flows)
A→B→C	Causal Chain	observed	Inactive (blocked) (A ‖ C|B)
A←B→C	Common Cause (fork)	Not observe	Active
A←B→C	Common Cause (fork)	observed	Inactive (blocked)
A→B←C	Common effect (collide)	Not observe	Inactive (A and C indep)
A→B←C	Common effect (collide)	B observed or any descendant of B is observed	Active

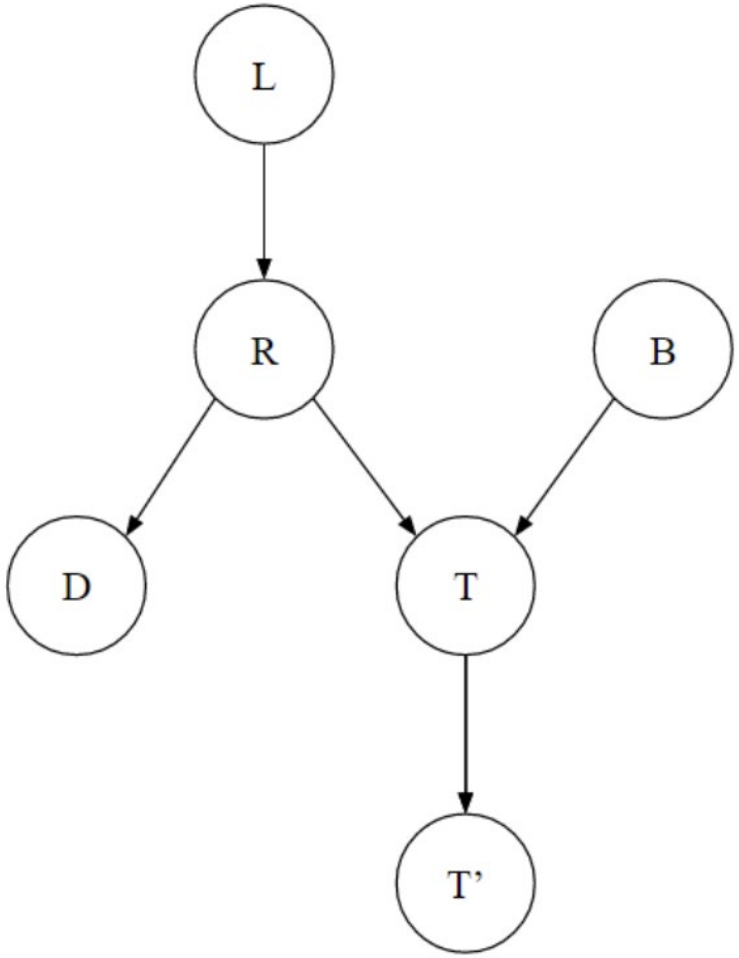
Bayesian Network

- Example
- Apply the d-separation algorithm if $X = L, Y = B$ and $Z = \{R, T\}$
  - Shade nodes R and T
  - Undirected paths from L to B: L – R – T – B
  - Decompose the path into overlapping triples
    - Triple 1: L → R → T. Since R is observed, this triple is blocked (inactive).
    - Triple 2: R → T ← B. Since T is observed, this triple is active.
  - Since a path is active only if every triple on it is active and Triple 1 is inactive, the path is inactive.
  - Since every path from L to B is inactive, L and B are d-separated from {R, T}.
  - Conclusion: L is conditionally independent of B given {R, T}.



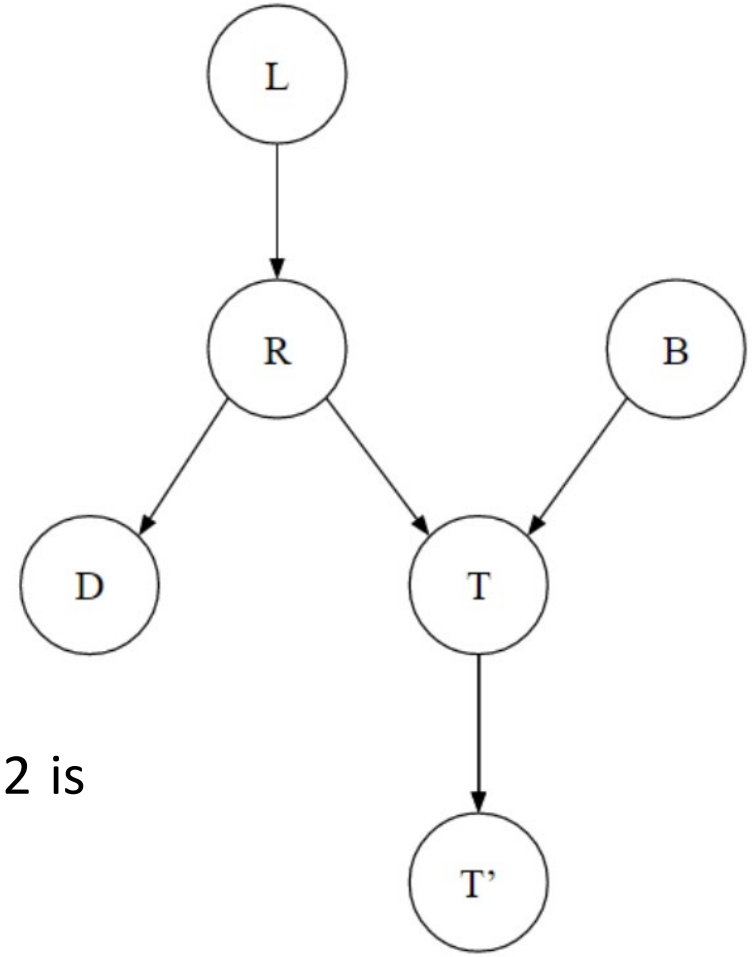
Bayesian Network

- Example
- Apply the d-separation algorithm if $X = L, Y = B$ and $Z = \{T'\}$
  - Shade node T'
  - Undirected paths from L to B: L – R – T – B
  - Decompose the path into overlapping triples
    - Triple 1: L → R → T. Since R is not observed, this triple is active.
    - Triple 2: R → T ← B. Since T' is observed, this triple is active.
  - Since a path is active only if every triple on it is active, the path is active.
  - Since there is a path from L to B is active, L and B are d-connected given T'.
  - Conclusion: The conditionally independent between L and B given T' is not guaranteed.



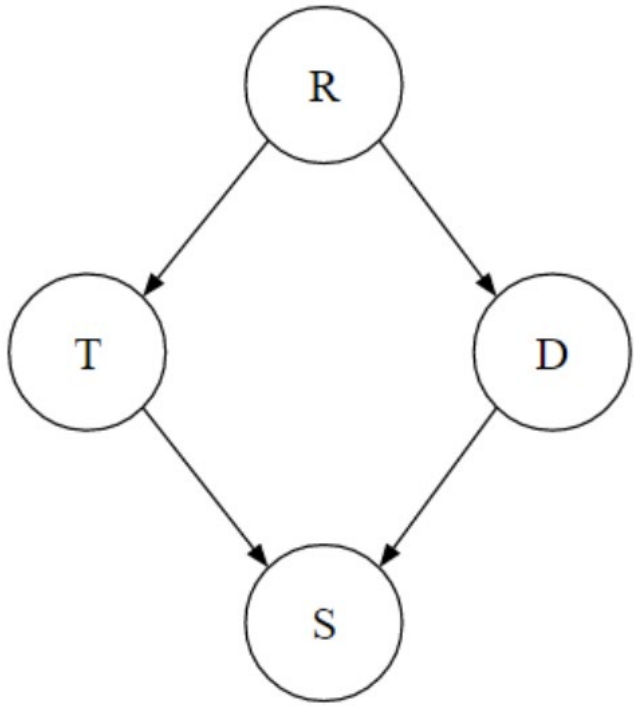
Bayesian Network

- Example
- Apply the d-separation algorithm if $X = L$, $Y = B$ and $Z = \emptyset$
  - No node is shaded.
  - Undirected paths from L to B: L – R – T – B
  - Decompose the path into overlapping triples
    - Triple 1: L → R → T. Since R is not observed, this triple is active.
    - Triple 2: R → T ← B. Since T and T' are not observed, this triple is inactive.
  - Since a path is active only if every triple on it is active and Triple 2 is inactive, the path is inactive.
  - Since every path from L to B is inactive, L and B are independent.



Bayesian Network

- Example
- Apply the d-separation algorithm if $X = T$, $Y = D$ and $Z = \{R, S\}$
  - Shade node R and S.
  - Undirected paths from T to D: T – S – D, T – R – D
  - Decompose the path into overlapping triples
    - Triple 1: T → S ← D. Since S is observed, this triple is active.
    - Triple 2: T ← R → D. Since R is observed, this triple is inactive.
  - Since a path is active only if every triple on it is active and Triple 2 is inactive, path 2 is inactive and path 1 is active.
  - Since there is a path from T to D is active, The conditionally independent between T and D given R and S is not guaranteed.



Bayesian Network

- Given a Bayesian network $P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$ and evidence $E = e$, where $E \subseteq X$, a subset of the set of variables.
- The objective is to calculate query conditional probability $P(Q = q|E = e)$ for all values of q, where $Q \subseteq X$ is the set of query.
- The four steps strategy:

1. Remove variables that are not ancestors of Q or E.
   - Non-ancestors cannot influence Q or E. They are irrelevant and can be safely removed.
   - This removes all nodes whose entire subtree cannot affect the query or the evidence.
2. Condition on the evidence $E = e$
   - Fix every evidence node to its observed value e
   - Replace the conditional probability table (CPT) of each evidence node's children with the reduced CPT that only uses the observed value.
   - The evidence nodes themselves can be removed after this step.

Bayesian Network

- The four steps strategy:

  3. Remove nodes now disconnected from Q.

     - After steps 1 – 2, some nodes may no longer have any path connecting them to any query variables in Q.

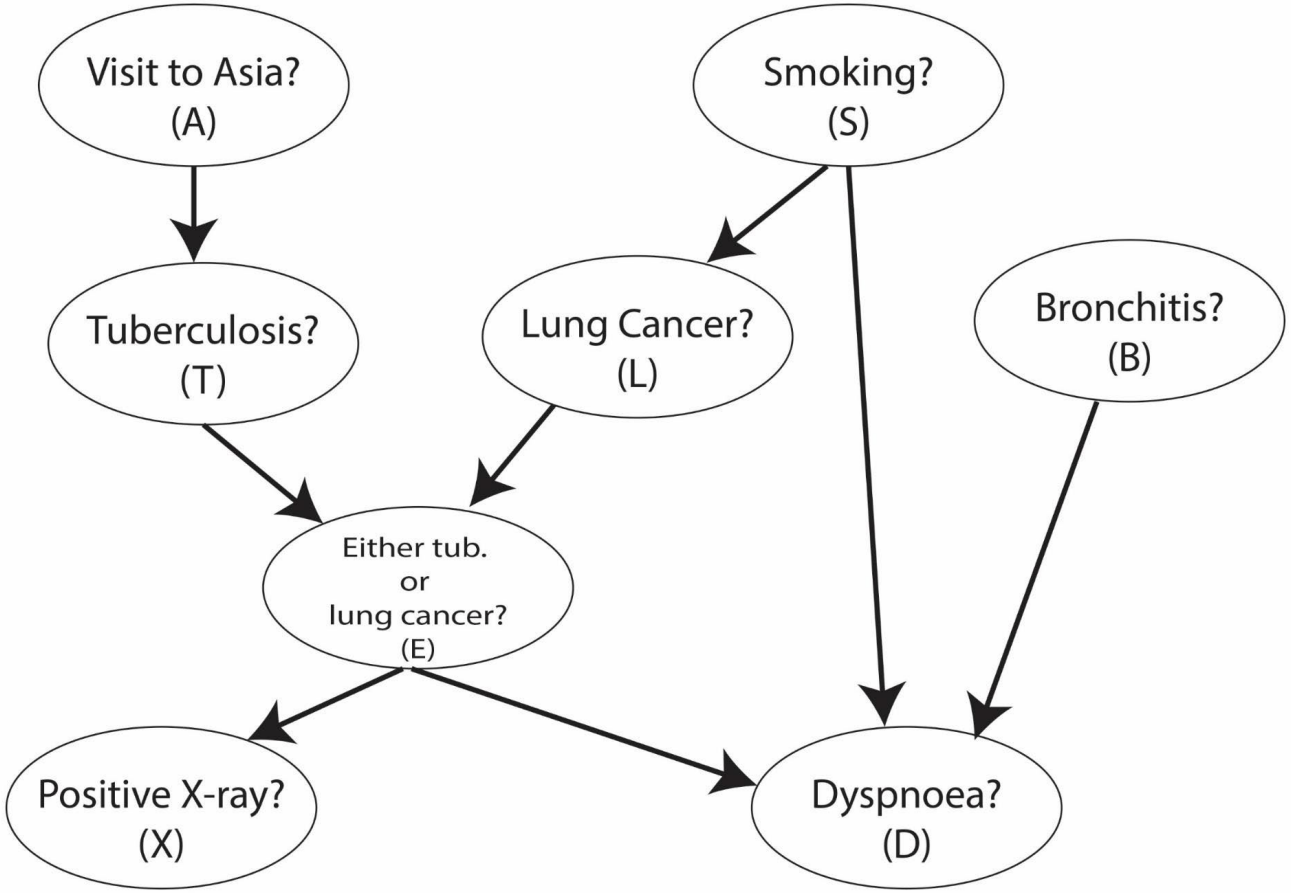
     - These nodes are now independent of Q given the evidence and can be removed.

  4. Evaluate the conditional probability on the reduced graph.

     - The remaining graph is much smaller and contains only the variables that matter.

Bayesian Network

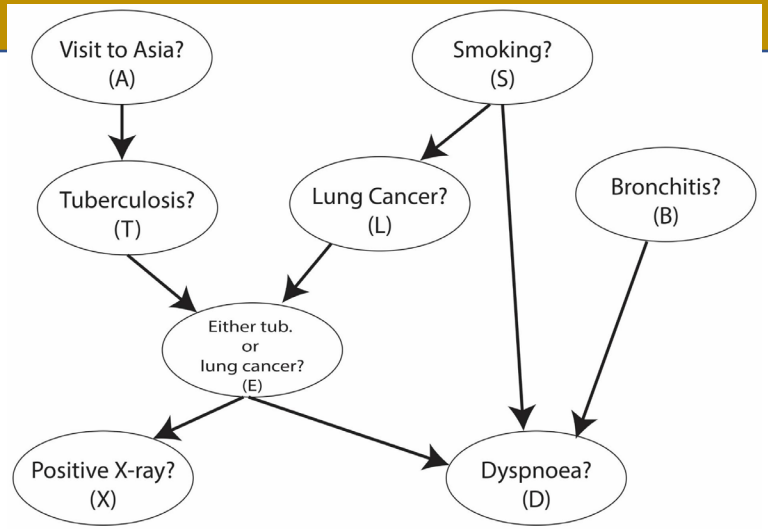
- Example Asia Network
- All binary variables (Yes=1, No=0)
  - A: Visit to Asis?
  - S: Smoking?
  - T: Tuberculosis?
  - L: Lung cancer?
  - E: either tuberculosis or lung cancer?
  - B: Bronchitis?
  - X: Positive X-ray?
  - D: Dyspnoea?



Bayesian Network

- Example Asia Network
- Standard CPT values
  - $P(A = 1) = 0.01$
  - $P(S = 1) = 0.5$
  - $P(T = 1|A = 1) = 0.05, P(T = 1|A = 0) = 0.01$
  - $P(L = 1|S = 1) = 0.10, P(L = 1|S = 0) = 0.01$
  - $P(E = 1|T = 1, L = 1) = 1,\ P(E = 1|T = 0, L = 1) = 1,\ P(E = 1|T = 1, L = 0) = 1,$ $P(E = 1|T = 0, L = 0) = 0$, else zero.
  - $P(X = 1|E = 1) = 0.98, P(X = 1|E = 0) = 0.05$
- Want to calculate $P(L = 1|X = 1)$

Bayesian Network



- Example Asia Network
- Step 1: Remove variables that are not ancestors of L or X.
  - Ancestors of L or X: A, T, L, E, S
  - Remove B and D.
- Condition on X=1
  - The CPT for X becomes a constant $P(X = 1|E = 1) = 0.98$ and $P(X = 1|E = 0) = 0.05$
  - X is removed.
- Remove nodes now disconnected from L
  - After fixing X and removing it, the remaining variables are A, S, T, L and E.
  - They are connected to L.
- Evaluate the conditional probability on the reduced graph
  - The reduced network has only 5 variables A, S, T, L, E

Bayesian Network

- Example Asia Network

$$P(L, X = 1) \propto \sum_{A,S,T,E} P(L, A, S, T, E, X = 1)$$

$$= \sum_{A,S,T,E} P(X = 1|E)P(E|T, L)P(L|S)P(T|A)P(A)\,P(S)$$

$$P(L = 1, X = 1) = P(X = 1|E = 1)P(E = 1|T = 1, L = 1)P(L = 1|S = 1)P(T = 1|A = 1)P(A = 1)P(S = 1) + P(X = 1|E = 1)P(E = 1|T = 1, L = 1)P(L = 1|S = 0)P(T = 1|A = 1)P(A = 1)P(S = 0) + P(X = 1|E = 1)P(E = 1|T = 1, L = 1)P(L = 1|S = 1)P(T = 1|A = 0)P(A = 0)P(S = 1) + P(X = 1|E = 1)P(E = 1|T = 1, L = 1)P(L = 1|S = 0)P(T = 1|A = 0)P(A = 0)P(S = 0) + P(X = 1|E = 1)P(E = 1|T = 0, L = 1)P(L = 1|S = 1)P(T = 0|A = 1)P(A = 1)P(S = 1) + P(X = 1|E = 1)P(E = 1|T = 0, L = 1)P(L = 1|S = 0)P(T = 0|A = 1)P(A = 1)P(S = 0) + P(X = 1|E = 1)P(E = 1|T = 0, L = 1)P(L = 1|S = 1)P(T = 0|A = 0)P(A = 0)P(S = 1) + P(X = 1|E = 1)P(E = 1|T = 0, L = 1)P(L = 1|S = 0)P(T = 0|A = 0)P(A = 0)P(S = 0) \approx 0.05390$$

Bayesian Network

- Example Asia Network
- Calculate similarly.

$$P(L = 0, X = 1) \approx 0.02833$$

- Hence,

$$P(L = 1|X = 1) = \frac{P(L = 1, X = 1)}{P(L = 0, X = 1) + P(L = 1, X = 1)} = 0.655$$

- Interpretation:
  - Prior probability of lung cancer was only about 5.5%. After observing a positive X-ray, it jumps to 65.5% — the X-ray is strong evidence for E, which in turn provides strong evidence for L (or T).